



UNIVERSITY OF ENGINEERING AND TECHNOLOGY MARDAN

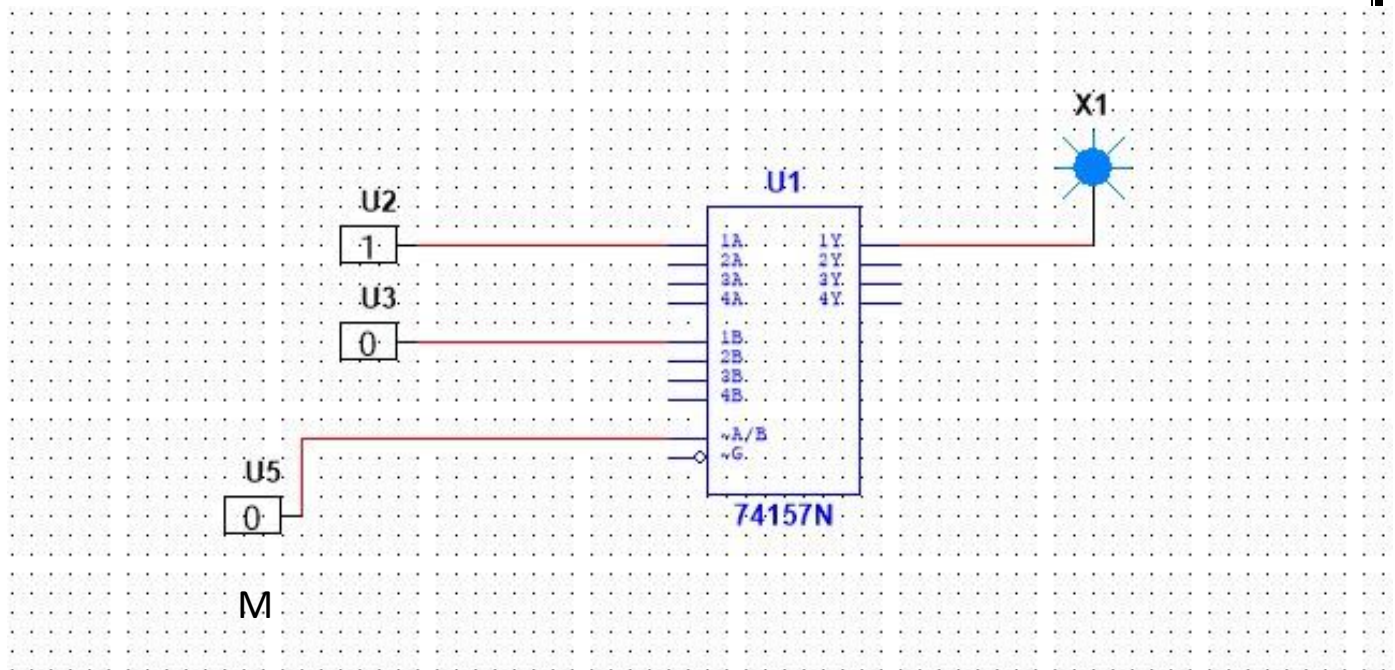
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REG NO	23MDBCS435
SECTION	A
DEPPT	COMPUTER SCIENCE
SUBJECT	DLD LAB
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LAB TASK : 11

Multiplexer

The multiplexer or data selector is a circuit that connects any one of several inputs to a single output. The circuit that performs the opposite function, in which a single input is directed to one of several outputs, is called a de-multiplexer or a decoder. These definitions are illustrated in Figure 11.1. The routing of the data is determined by additional logic signals called the select or (or address) inputs.

Circuit Diagram:



U5=0 ,A will be selected

U5=1, B will be selected

TRUTH TABLE



INPUT			OUTPUTS
SELECT A/B	A	B	
0	0	x	0
0	1	x	1
1	x	0	0
1	x	1	1

FUNCTION FOR THE CIRCUIT:

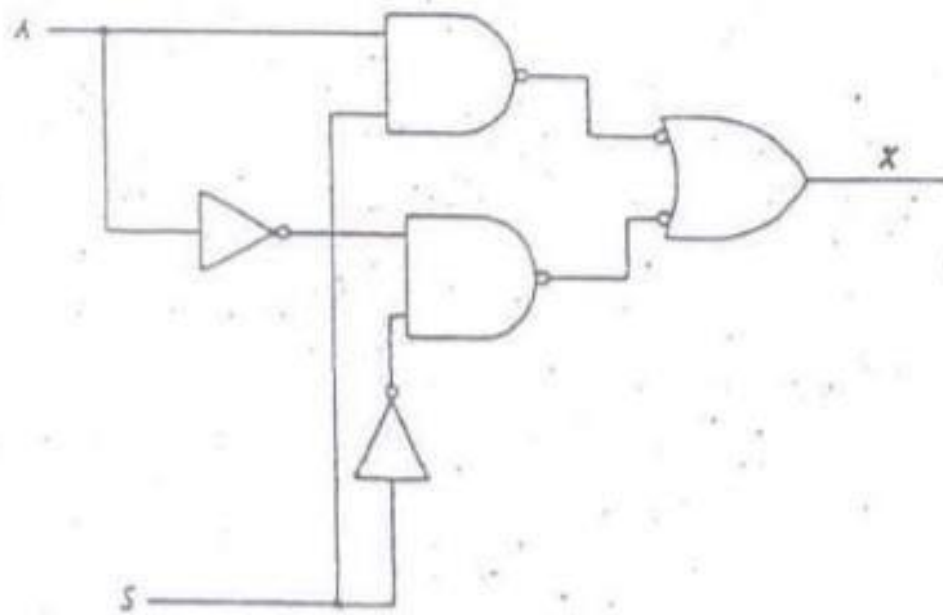


Figure 11.4

we have the following components:

1. An inverter (NOT gate).
2. An AND gate.
3. An OR gate.
4. A signal labeled 'S'.

Putting it all together:

$$X = B \vee S$$

$$X = (A \wedge S) \vee S$$

Now, let's simplify further:

$$X = A \vee S$$

Therefore, the derived function for this circuit is:

$$F(A, S)_n = AVS$$

THE END